

1. Run nslookup to obtain the IP address of the web server for the Indian Institute of Technology in Bombay, India: www.iitb.ac.in. What is the IP address of www.iitb.ac.in

-> 103.21.124.133

2. What is the IP address of the DNS server that provided the answer to your nslookup command in question 1 above?

-> 127.0.0.53

3. Did the answer to your nslookup command in question 1 above come from an authoritative or non-authoritative server?

-> Non-authoritative answer:

4. Use the nslookup command to determine the name of the authoritative name server for the iit.ac.in domain. What is that name? (If there are more than one authoritative servers, what is the name of the first authoritative server returned by nslookup)? If you had to find the IP address of that authoritative name server, how would you do so?

-> Authoritative name servers for iitb.ac.in:

[dns3.iitb.ac.in](#) (first)

dns1.iitb.ac.in

[dns2.iitb.ac.in](#)

```
student@lab707-29:~$ nslookup dns3.iitb.ac.in i will use this
```

```
Server:          127.0.0.53
```

```
Address: 127.0.0.53#53
```

```
Non-authoritative answer:
```

```
Name: dns3.iitb.ac.in
```

```
Address: 103.21.127.129
```

```
student@lab707-29:~$
```

5. Locate the first DNS query message resolving the name `gaia.cs.umass.edu`. What is the packet number in the trace for the DNS query message? Is this query message sent over UDP or TCP?

-> 15, UDP

6. Now locate the corresponding DNS response to the initial DNS query. What is the packet number in the trace for the DNS response message? Is this response message received via UDP or TCP?

-> 17, UDP

7. What is the destination port for the DNS query message? What is the source port of the DNS response message?

-> 53, 58350

8. To what IP address is the DNS query message sent?

-> 75.75.75.75

9. Examine the DNS query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?

-> questions 1, answer 0

10. Examine the DNS response message to the initial query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?

-> questions 1, answer 1

11. The web page for the base file http://gaia.cs.umass.edu/kurose_ross/ references the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E_2.jpg , which, like the base webpage, is on gaia.cs.umass.edu. What is the packet number in the trace for the initial HTTP GET request for the base file http://gaia.cs.umass.edu/kurose_ross/?

What is the packet number in the trace of the DNS query made to resolve gaia.cs.umass.edu so that this initial HTTP request can be sent to the gaia.cs.umass.edu IP address?

What is the packet number in the trace of the received DNS response?

What is the packet number in the trace for the HTTP GET request for the image object

http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_8E2.jpg?

What is the packet number in the DNS query made to resolve gaia.cs.umass.edu so that this second HTTP request can be sent to the gaia.cs.umass.edu IP address?

Discuss how DNS caching affects the answer to this last question.

-> 22, 15, 17, 205

12. What is the destination port for the DNS query message? What is the source port

of the DNS response message?

src port:57837 , dst port:53

13. To what IP address is the DNS query message sent? Is this the IP address of your

default local DNS server?

src:10.0.0.44 dest:75.75.75.75, yes

14. Examine the DNS query message. What "Type" of DNS query is it? Does the

query message contain any "answers"?

Query type:A , answer:0

15. Examine the DNS response message to the query message. How many

"questions" does this DNS response message contain? How many "answers"?

Questions:1

Answers RR:1

16. To what IP address is the DNS query message sent? Is this the IP address of your

default local DNS server?

Sent to IP:75.75.75.75, yes

17. Examine the DNS query message. How many questions does the query have?

Does the query message contain any "answers"?

Questions:1

Answer:0

18. Examine the DNS response message (in particular the DNS response message that

has type "NS"). How many answers does the response have? What information

is contained in the answers? How many additional resource records are returned?

Answers:3

Information: ns1.umass.edu, ns3.umass.edu, ns2.umass.edu

Additional RRs: 128.119.10.28, 128.119.10.27, 128.103.38.68